

Project Phase 6 — Loan Approval Prediction

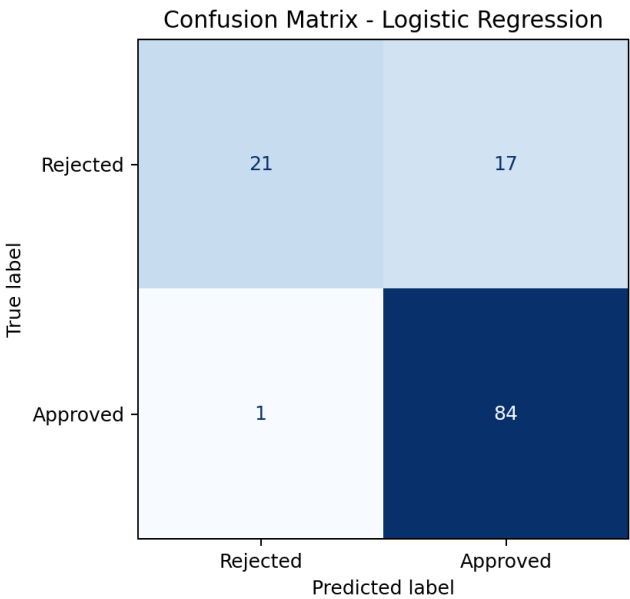
Model Evaluation — Logistic Regression (Classification Project)

This is a classification project (Loan_Status: Approved/Rejected), so the model is evaluated using classification metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC — evaluated on the 123-row held-out test set.

1. All Evaluation Metrics

Metric	Value	What it means
Accuracy	85.37%	Overall % of correct predictions
Precision	83.17%	Of predicted approvals, % that were correct
Recall (Sensitivity)	98.82%	Of actual approvals, % correctly identified
F1-Score	90.32%	Harmonic mean of Precision & Recall
ROC-AUC	0.8536	Model's ability to rank approvals above rejections

2. Confusion Matrix



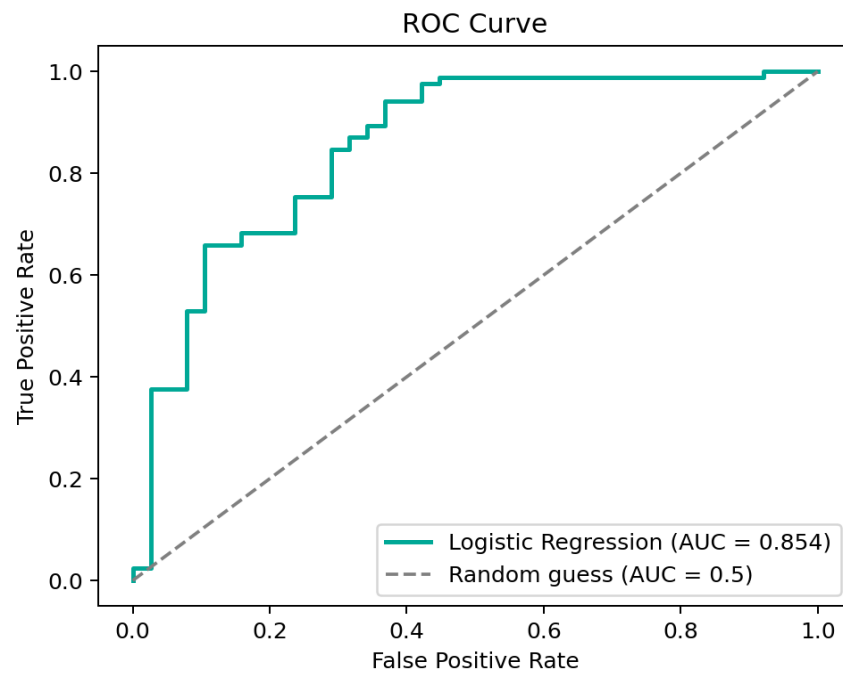
	Predicted: Rejected	Predicted: Approved
Actual: Rejected	21 (True Negative)	17 (False Positive)
Actual: Approved	1 (False Negative)	84 (True Positive)

21 rejections and 84 approvals were correctly predicted. 17 applicants who were actually rejected were incorrectly predicted as approved (false positives), while only 1 actually-approved applicant was incorrectly predicted as rejected (false negative).

3. Classification Report

Class	Precision	Recall	F1-Score	Support
Rejected (0)	0.9545	0.5526	0.7000	38
Approved (1)	0.8317	0.9882	0.9032	85
Accuracy			0.8537	123
Macro Avg	0.8931	0.7704	0.8016	123
Weighted Avg	0.8696	0.8537	0.8404	123

4. ROC Curve



ROC-AUC = 0.854, meaning the model has an 85.4% chance of ranking a randomly chosen approved applicant higher (more likely to be approved) than a randomly chosen rejected applicant — well above the 0.5 baseline of random guessing.

5. Model Analysis

- Overall performance is strong for a simple linear model: 85.4% accuracy and 0.854 ROC-AUC on unseen test data.
- The model is much better at identifying Approved loans (Recall 98.8%, Precision 83.2%) than Rejected loans (Recall 55.3%, Precision 95.4%) — it rarely misses a true approval, but it does approve a notable number of applicants who were actually rejected.
- This asymmetry directly reflects the class imbalance in the data (~69% Approved vs ~31% Rejected) and the dominance of Credit_History as a predictor: applicants with good credit history are almost always predicted 'Approved', even when other factors would normally lead to rejection.
- For a real lending use-case, this behavior is risky (high false-positive rate on approvals) — techniques like class-weighting (class_weight='balanced'), adjusting the decision threshold, or adding more discriminating features could improve Rejected-class recall.
- Given F1-score (90.3%) and ROC-AUC (0.854) are both comfortably above the 0.5-baseline, the Logistic Regression model has learned a genuine, non-trivial relationship between applicant features and loan approval, not random noise.